1 FIG-005 — Mace 1ns overleaf B to G — seed-aware diffusion benchmark

Group: MD-1NS **Evidence chain:** seed-aware audited **Status:** publication candidate; REV-C9 partial

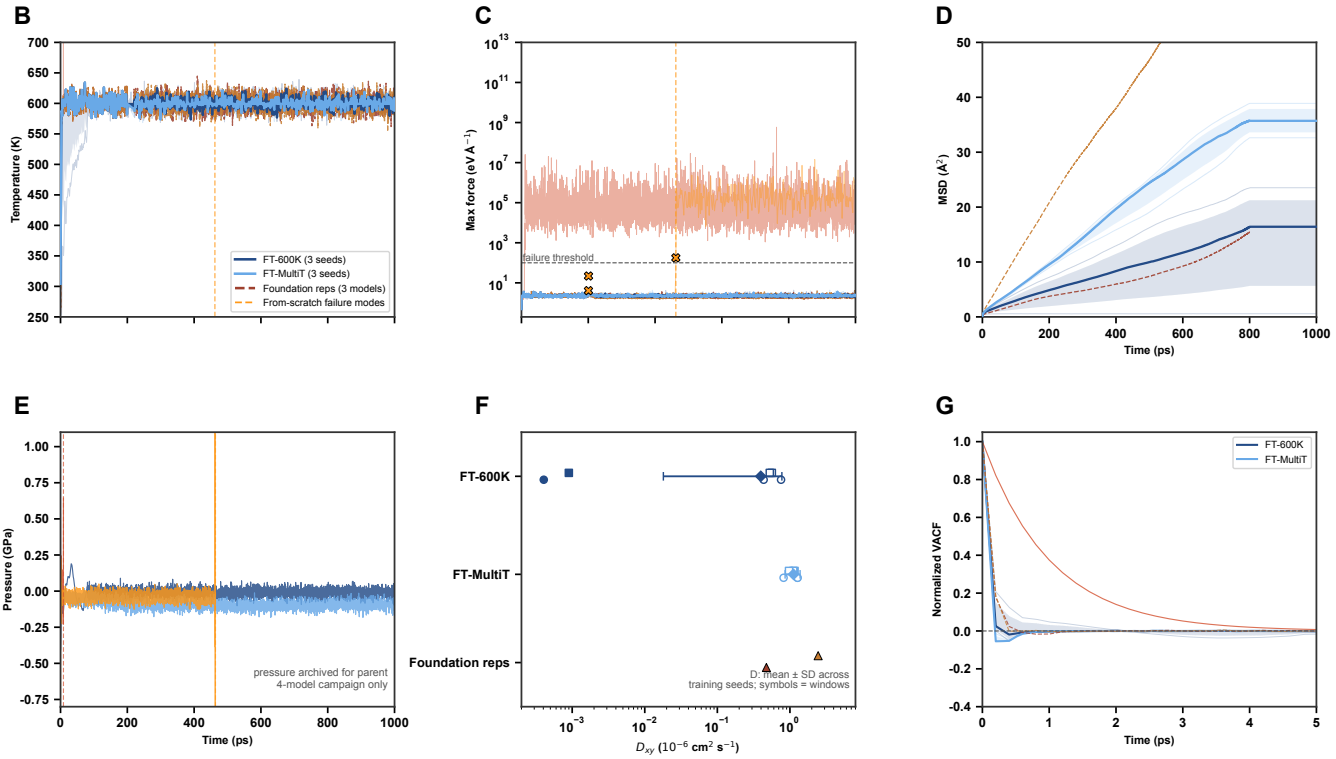


Figure 1: **Seed-aware one-nanosecond molecular-dynamics benchmark.** (B) Temperature traces for the stable FT-600K and FT-MultiT families, summarized across training seeds 123/234/345, with the Foundation and From-scratch failure branches retained as explicit diagnostics. (C) Maximum-force trajectories and scientific-threshold markers used to identify instability. (D) Multi-origin MSD with seed-aware spread for the stable FT families and representative foundation trajectories; the From-scratch branches are shown as failure modes rather than transport estimates. (E) Pressure availability: the archived pressure trace is restricted to the parent four-model campaign. (F) Window-specific D_{xy} estimates with mean ± SD across training seeds where applicable. (G) Normalized VACF for the stable FT families. No HFACF or thermal-conductivity κ is reported because no same-campaign heat-flux time series was archived.

Results text draft. The updated panels summarize the completed 1 ns audit for three FT-600K training seeds, three FT-MultiT training seeds, three foundation representatives, and three From-scratch failure-mode seeds. All MD runs use velocity seed 1100; therefore the three-seed spreads quantify training-seed uncertainty, not independent velocity-replica uncertainty. FT-600K gives $D_{xy} = (3.965 \pm 3.784) \times 10^{-7}$ cm^2 s^{-1} in the 200–500 ps window and $(3.640 \pm 3.149) \times 10^{-7}$ cm^2 s^{-1} in the 500–900 ps window. FT-MultiT gives $(1.120 \pm 0.261) \times 10^{-6}$ and $(1.080 \pm 0.097) \times 10^{-6}$ cm^2 s^{-1} in the same windows. From-scratch seeds 123, 234, and 345 fail the scientific stability gate at approximately 220.0, 200.8, and 200.6 ps, respectively, and are reported as *unstable — no D_{xy} reported*. **Discussion and boundary.** This figure supports the revised dynamic-stability and diffusion discussion and directly documents the From-scratch failure mode instead of silently dropping unstable trajectories. The FT uncertainty bands are across training seeds, while the common MD velocity seed means they are not independent velocity replicas. The figure must not be used to imply heat-flux-derived thermal conductivity: HFACF and κ remain unavailable because the audited 1 ns campaign did not archive a same-campaign heat-flux series. Foundation-medium-mpa-0 also shows a window-disagreement caution and should not be collapsed into a single overconfident transport number. **Article placement.** Results: dynamic stability, failure-mode benchmark, and diffusion; SI: seed-aware uncertainty and audit details. Do not place this figure under a thermal-conductivity claim unless a separate heat-flux campaign is added. **Reviewer linkage:** overall status PARTIAL.

- **REV-C9 (PARTIAL)**: The corrected matched-protocol FT-600K and FT-MultiT 1 ns trajectories now include training seeds 123/234/345, block-level D_{xy} estimates, and seed-aware uncertainty; From-scratch seeds are retained as audited failure modes. The remaining boundary is explicit: no same-campaign heat-flux series was archived, so HFACF/κ cannot be reconstructed or claimed from these trajectories. A dedicated heat-flux campaign would be required to close the thermal-conductivity portion of C9.

Evidence pointers.

- Seed-aware results: main/analysis/mace_1ns_seed_aware_20261011/seed_aware_results.csv
- Seed-aware uncertainty: main/analysis/mace_1ns_seed_aware_20261011/seed_aware_uncertainty.csv
- Seed-aware figure manifest: main/analysis/mace_1ns_seed_aware_20261011/figure_manifest.json
- Data/processed: main/analysis/mace_1ns_publication_figure_20261005/figure_manifest.json
- Data/processed: main/analysis/mace_1ns_publication_figure_20261005/final_benchmark_manifest.json
- Plot/processing script: main/analysis/mace_1ns_seed_aware_20261011/plot_seed_aware_final.py
- Plot/processing script: main/analysis/mace_1ns_publication_figure_20261005/plot_figure.py
- Plot/processing script: main/analysis/mace_1ns_publication_figure_20261005/plot_final_benchmark.py
- Caption/text source: main/analysis/mace_1ns_seed_aware_20261011/RESULTS_DISCUSSION.md
- Caption/text source: main/analysis/mace_1ns_seed_aware_20261011/figure_caption.md
- Caption/text source: main/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.md

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/figure_caption.md
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.md
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.tex
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/strict_replica_caption.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/figure_manifest.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_manifest.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/overleaf_panels_manifest.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/strict_replica_manifest.json
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_figure.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_final_benchmark.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_overleaf_panels.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_strict_replica.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_lag_remote.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_msd_fast.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_vacf_fast.py

Audit boundary: Multi-seed diffusion and From-scratch failure-mode closure are complete; HFACF/κ remain unavailable without a same-campaign heat-flux trajectory.